Notes on hyperbolic 5- and 6-simplex volumes formulas, reductions, and a practical code path

Abstract

This note records a workable extension of the earlier $\mathbb{H}^4$ discussion to compact simplices in $\mathbb{H}^5$ and $\mathbb{H}^6$. In dimension 5 the natural tool is Schlaefli's differential formula, which reduces the problem to hyperbolic tetrahedral 3-faces. In dimension 6 the even-dimensional Poincare-Schlaefli-Hopf reduction gives an absolute volume formula in terms of the 21 dihedral angles, the 35 spherical tetrahedral links of the 2-faces, and the 7 spherical 5-simplex links of the vertices. We also explain the numerical strategy used in the companion Python module: QMC evaluation of Klein-model simplex integrals, intrinsic extraction of lower-dimensional Klein coordinates from ambient hyperboloid Gram data, and gnomonic evaluation of spherical link volumes.

1 General simplex data from Klein vertices

Let

$$\Delta = [x_0, \dots, x_n] \subset B^n = \{x \in \mathbb{R}^n : \|x\| < 1\}$$

be a compact hyperbolic n-simplex in the Klein ball. Its hyperboloid lifts are

$$h_i = \frac{(1, x_i)}{\sqrt{1 - \|x_i\|^2}} \in \mathbb{H}^n \subset \mathbb{R}^{n,1}, \qquad \langle u, v \rangle = -u_0 v_0 + u_1 v_1 + \cdots + u_n v_n.$$

The unnormalized Klein matrix is

$$Q = (Q_{ij}), \qquad Q_{ij} = 1 - x_i \cdot x_j.$$

Since the true vertex Gram matrix is

$$G_{ij} = -\langle h_i, h_j \rangle = \frac{1 - x_i \cdot x_j}{\sqrt{(1 - \|x_i\|^2)(1 - \|x_j\|^2)}},$$

normalization cancels in the inverse, and if $R = Q^{-1}$ then the angle Gram matrix is

$$A_{ii} = 1, \qquad A_{ij} = -\frac{R_{ij}}{\sqrt{(-R_{ii})(-R_{jj})}} \quad (i \neq j). \tag{1}$$

Hence the dihedral angle between the facets opposite x_i and x_j is

$$\theta_{ij} = \arccos(-A_{ij}). \tag{2}$$

Formula (1) is the same linear-algebra package already used in the $\mathbb{H}^4$ note; it is just dimension-independent.

For numerical cross-checks it is useful to remember the direct Klein integral

$$\mathrm{Vol}_n(\Delta) = \int_{\Delta_E} \frac{dx}{(1 - \|x\|^2)^{(n+1)/2}}, \tag{3}$$

where $\Delta_E = \mathrm{conv}(x_0, \dots, x_n) \subset \mathbb{R}^n$ is the ordinary Euclidean simplex in the Klein ball. The companion code evaluates (3) by Sobol QMC sampling on the standard simplex.

2 The $\mathbb{H}^5$ case

Schlaefli differential formula

For a smoothly varying family of compact hyperbolic 5-polytopes, Schlaefli's differential formula reads

$$d\,\mathrm{Vol}_5(\Delta) = -\frac{1}{4}\sum_{F^3}\mathrm{Vol}_3(F^3)\,d\theta_F, \tag{4}$$

where the sum runs over the codimension-2 faces, i.e. the 3-dimensional tetrahedral faces. For a 5-simplex there are $\binom{6}{2} = 15$ such faces. This is the correct replacement for the $\mathbb{H}^4$ "absolute" reduction: in odd dimension one no longer gets volume directly from the alternating angle sum, so one integrates Schlaefli instead [3, 1].

If the simplex is

$$\Delta = [x_0, \ldots, x_5] \subset B^5,$$

set

$$\Delta_t = [tx_0, \ldots, tx_5], \qquad 0 < t \le 1.$$

Then Δ_t stays compact and collapses to the origin as $t \to 0$. Choosing some small $t_0 \in (0, 1)$, one has

$$\mathrm{Vol}_5(\Delta) = \mathrm{Vol}_5(\Delta_{t_0}) - \frac{1}{4}\int_{t_0}^{1}\sum_{i<j}\mathrm{Vol}_3(F_{ij}(t))\,d\theta_{ij}(t), \tag{5}$$

where $F_{ij}(t)$ is the tetrahedral face spanned by the four vertices with indices different from i, j.

Extracting the intrinsic $\mathbb{H}^3$ face geometry

The code does not implement the hyperbolic tetrahedron dilogarithm formula explicitly, although that would also be possible using Murakami–Yano/Ushijima. Instead it extracts intrinsic Klein coordinates for each tetrahedral face and applies the direct 3-dimensional Klein integral.

Suppose $u_0, \ldots, u_3 \in B^5$ are the four Klein vertices of a tetrahedral face. Lift them to the hyperboloid as above, and form the vertex Gram matrix

$$G^{\mathrm{face}}_{ab} = -\langle h_a, h_b\rangle, \qquad 0 \le a, b \le 3.$$

Equivalently, the Minkowski inner-product matrix of the lifted vertices is

$$M_{ab} = \langle h_a, h_b\rangle.$$

Because the face is intrinsically 3-dimensional, M has signature $(1, 3)$. Numerically one factors it as

$$M = C^T J_{3,1} C, \qquad J_{3,1} = \mathrm{diag}(-1, 1, 1, 1),$$

so that the columns of C are hyperboloid coordinates

$$y_a = (y_{a,0}, y_{a,1}, y_{a,2}, y_{a,3}) \in \mathbb{H}^3.$$

Then the intrinsic Klein coordinates of the face are simply

$$z_a = \frac{(y_{a,1}, y_{a,2}, y_{a,3})}{y_{a,0}} \in B^3. \tag{6}$$

Finally,

$$\mathrm{Vol}_3(F) = \int_{\mathrm{conv}(z_0,z_1,z_2,z_3)} \frac{dz}{(1-\|z\|^2)^2}. \tag{7}$$

Combining (5) and (7) gives a completely explicit numerical pipeline from the original five-dimensional Klein vertices.

Remark 1. *At the theoretical level, hyperbolic tetrahedron volumes are already dilogarithmic, and Kellerhals showed that in hyperbolic* 5*-space the general odd-dimensional picture reaches trilogarithms, with doubly asymptotic* 5*-orthoschemes providing a fundamental building block [4, 6, 7]. The code in this note stays closer to the raw simplex data and uses QMC for the lower-dimensional face integrals instead.*

3 The $\mathbb{H}^6$ case

The even-dimensional Hopf reduction

Let w_k denote the sum of the normalized $(6-k)$-fold interior angles at the k-faces of a 6-simplex. Thus $w_6 = 1$; w_4 is built from the 21 codimension-2 dihedral angles; w_2 comes from the 35 spherical tetrahedral links of the 2-faces; and w_0 comes from the 7 spherical 5-simplex vertex links.

The Peschl relations imply

$$W(\Delta^6) = w_0 - \frac{1}{2}w_2 + w_4 - \frac{17}{4}, \tag{8}$$

while the hyperbolic Poincare–Schlaefli–Hopf theorem gives

$$W(\Delta^6) = -\frac{2}{\Omega_6}\,\mathrm{Vol}_6(\Delta^6), \qquad \Omega_6 = \mathrm{Vol}(\mathbb{S}^6) = \frac{16\pi^3}{15}. \tag{9}$$

Therefore

$$\boxed{\mathrm{Vol}_6(\Delta) = \frac{8\pi^3}{15}\left(\frac{17}{4} - w_4 + \frac{1}{2}w_2 - w_0\right).} \tag{10}$$

This is the 6-dimensional analogue of the $\mathbb{H}^4$ formula, except that now the unresolved term is an $\mathbb{S}^5$-volume rather than an $\mathbb{S}^3$-volume [2, 1].

Writing the three pieces from the angle Gram matrix

Let

$$\Delta = [x_0, \ldots, x_6] \subset B^6,$$

and let A be its angle Gram matrix from (1). Then:

The w_4 term. The codimension-2 faces correspond to pairs $\{i, j\}$, so

$$w_4 = \frac{1}{2\pi} \sum_{0 \le i < j \le 6} \theta_{ij}. \tag{11}$$

The w_2 term. A 2-face is the intersection of four facets, hence corresponds to a 4-subset $I \subset \{0, \dots, 6\}$. The spherical tetrahedral link at that face has angle Gram matrix equal to the principal submatrix A_I. Therefore

$$w_2 = \frac{1}{2\pi^2} \sum_{|I|=4} \mathrm{Vol}_{\mathbb{S}^3}(L_I), \qquad \text{with } \mathrm{Gram}(L_I) = A_I. \tag{12}$$

Each $\mathrm{Vol}_{\mathbb{S}^3}(L_I)$ is computable by Murakami's explicit logarithm/dilogarithm formula. This is exactly the same special-function kernel that already appeared in the $\mathbb{H}^4$ vertex-link term [5].

The w_0 term. The vertex link at x_i is the spherical 5-simplex obtained by deleting row and column i:

$$w_0 = \frac{1}{\pi^3} \sum_{i=0}^{6} \mathrm{Vol}_{\mathbb{S}^5}(L_i), \qquad \text{with } \mathrm{Gram}(L_i) = A^{(i)}. \tag{13}$$

If one passes to generalized simplices and a vertex becomes ideal, then the corresponding term in w_0 vanishes by continuity. In particular, for an ideal 6-simplex one gets the simplified formula

$$\mathrm{Vol}_6(\Delta_{\text{ideal}}) = \frac{8\pi^3}{15} \left(\frac{17}{4} - w_4 + \frac{1}{2} w_2 \right). \tag{14}$$

How the code evaluates the $\mathbb{S}^5$ links

Let B be a 6×6 spherical angle Gram matrix. The associated spherical vertex Gram matrix is

$$G_{ab} = \frac{(B^{-1})_{ab}}{\sqrt{(B^{-1})_{aa}(B^{-1})_{bb}}}. \tag{15}$$

Choose vertices $v_0, \dots, v_5 \in \mathbb{S}^5 \subset \mathbb{R}^6$ with $v_a \cdot v_b = G_{ab}$. To put the simplex into gnomonic coordinates, solve

$$G\,\alpha = \mathbf{1}, \qquad c' = \sum_{a=0}^{5} \alpha_a v_a, \qquad c = \frac{c'}{\|c'\|} \in \mathbb{S}^5.$$

Then c lies in the interior hemisphere of the simplex. If $e_1, \dots, e_5$ is an orthonormal basis of $c^\perp$, the gnomonic coordinates are

$$x_a = \frac{1}{v_a \cdot c} \big((v_a \cdot e_1), \dots, (v_a \cdot e_5) \big) \in \mathbb{R}^5. \tag{16}$$

Under the inverse gnomonic map

$$x \longmapsto \frac{c + \sum_{r=1}^{5} x_r e_r}{\sqrt{1 + \|x\|^2}},$$

the spherical volume element becomes

$$d\,\mathrm{Vol}_{\mathbb{S}^5} = \frac{dx}{(1 + \|x\|^2)^3}. \tag{17}$$

Hence the code evaluates each vertex link by

$$\mathrm{Vol}_{\mathbb{S}^5}(L_i) = \int_{\mathrm{conv}(x_0, \dots, x_5)} \frac{dx}{(1 + \|x\|^2)^3}, \tag{18}$$

again using Sobol QMC on the Euclidean simplex in gnomonic coordinates.

4 Practical remarks and what transfers to polytopes

1. For $\mathbb{H}^5$, the code's main routine is the Schlaefli path integrator (5). A direct 5-dimensional Klein integral is included as a cross-check.
2. For $\mathbb{H}^6$, the code's main routine is the absolute reduction formula (10). A direct 6-dimensional Klein integral is again included as a cross-check. Near the Euclidean or very-small-volume regime, (10) subtracts several large angle-sum terms and can suffer noticeable cancellation. In that regime the direct Klein integral is the more reliable diagnostic.
3. For general polytopes one may triangulate and sum simplex volumes. In even dimension the Hopf face-sum formula also extends directly to a polytope, but from the coding perspective triangulation is usually the simplest path.

5 Two numerical smoke tests

With the defaults in the companion file `hyperbolic_5_6_simplex_volume.py`, the following examples are useful quick checks.

Space	simplex	main routine	direct Klein cross-check
$\mathbb{H}^5$	$\mathrm{conv}(0, 0.5e_1, \ldots, 0.5e_5)$	3.14018×10^{-4}	3.14365×10^{-4}
$\mathbb{H}^6$	$\mathrm{conv}(0, 0.7e_1, \ldots, 0.7e_6)$	2.44670×10^{-4}	2.44975×10^{-4}

The point of the second row is not that the reduction formula is "more exact" numerically, but rather that its constituent lower-dimensional pieces are now explicit enough to make an implementation practical. The direct integral remains the best sanity check.

References

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